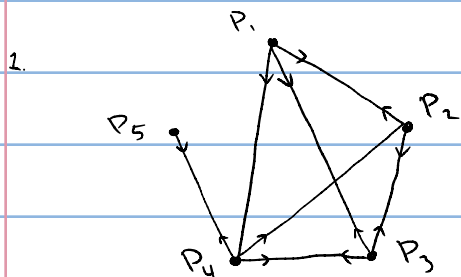


Homework 3 Key

Solutions by:
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$$\begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$2. A_1^2 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 0 & 1 \end{bmatrix}$$

$$A_2^2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

$$A_1^3 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 7 \\ 0 & 1 \end{bmatrix}$$

$$A_2^3 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$$

$$\vdots$$

$$A_1^n = \begin{bmatrix} 2^n & 2^n - 1 \\ 0 & 1 \end{bmatrix}$$

$$\vdots$$

$$A_2^n = \begin{bmatrix} 2^{n-1} & 2^{n-1} \\ 2^{n-1} & 2^{n-1} \end{bmatrix}$$

$$A_3^2 = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a^2 & ab \\ 0 & 0 \end{bmatrix}$$

$$A_3^3 = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a^2 & ab \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a^3 & a^2b \\ 0 & 0 \end{bmatrix}$$

$$\vdots$$

$$A_3^n = \begin{bmatrix} a^n & a^{n-1}b \\ 0 & 0 \end{bmatrix}$$

3. Let $A = (a_{ij})$ and $B = (b_{ij})$

$$(A - B)^T = (a_{ij} - b_{ij})^T = (a_{ji} - b_{ji}) = (a_{ji}) - (b_{ji}) = (a_{ij})^T - (b_{ij})^T = A^T - B^T$$

4. a. $\text{tr}(cA) = \sum_{i=1}^n ca_{ii} = c \sum_{i=1}^n a_{ii} = c \text{tr}(A)$

b. $\text{tr}(A+B) = \sum_{i=1}^n (a_{ii} + b_{ii}) = \sum_{i=1}^n a_{ii} + \sum_{i=1}^n b_{ii} = \text{tr}(A) + \text{tr}(B)$

$$c. AB = (c_{ij}) \quad c_{ii} = a_{i1}b_{1i} + \dots + a_{in}b_{ni} = \sum_{j=1}^n a_{ij}b_{ji}$$

$$BA = (d_{ij}) \quad d_{jj} = b_{j1}a_{1j} + \dots + b_{jn}a_{nj} = \sum_{i=1}^n b_{ji}a_{ij}$$

$$\text{tr}(AB) = \sum_{i=1}^n \sum_{j=1}^n a_{ij}b_{ji} = \sum_{j=1}^n \sum_{i=1}^n b_{ji}a_{ij} = \text{tr}(BA)$$

$$d. \text{tr}(A^T) = \sum_{i=1}^n a_{ii} = \text{tr}(A)$$

$$e. A^T A = (c_{ij}) \quad c_{ii} = a_{i1}a_{1i} + \dots + a_{ni}a_{ni} = \sum_{j=1}^n a_{ji}^2$$

$$\text{tr}(A^T A) = \sum_{i=1}^n \sum_{j=1}^n a_{ji}^2 \geq 0$$

$$s. C = AB$$

$$CC^{-1} = ABC^{-1}$$

$$I = ABC^{-1}$$

$$A^{-1} = A^{-1}ABC^{-1}$$

$$A^{-1} = BC^{-1}$$

$$6.a. 3 \begin{vmatrix} -1 & 5 \\ 3 & 2 \end{vmatrix} - 2 \begin{vmatrix} 1 & 2 \\ 3 & 2 \end{vmatrix} = 3[(-1)(2) - (3)(5)] - 2[(1)(2) - (3)(2)]$$

$$= 3(-2-15) - 2(2-6)$$

$$= -51 + 8$$

$$= -43$$

$$1.b. 4 \begin{vmatrix} 2 & 0 & 3 \\ 0 & 3 & 4 \\ -3 & 2 & 1 \end{vmatrix} - 1 \begin{vmatrix} -4 & 2 & 1 \\ 0 & 3 & 4 \\ -3 & 2 & 1 \end{vmatrix} + 2 \begin{vmatrix} -4 & 2 & 1 \\ 2 & 0 & 3 \\ -3 & 2 & 1 \end{vmatrix}$$

$$= 4 \begin{vmatrix} 2 & 3 & 4 \\ 2 & 1 & -3 \end{vmatrix} + 3 \begin{vmatrix} 0 & 3 \\ -3 & 2 \end{vmatrix} - \begin{vmatrix} -4 & 3 & 4 \\ 2 & 1 & 3 \end{vmatrix} - 3 \begin{vmatrix} 2 & 1 \\ 3 & 4 \end{vmatrix} + 2 \begin{vmatrix} -2 & 2 & -3 \\ 2 & 1 & -3 \end{vmatrix} + 2 \begin{vmatrix} -4 & 2 \\ -3 & 2 \end{vmatrix}$$

$$= 8(3-8) + 12(0+9) + 4(3-8) + 3(8-3) - 4(2-2) - 6(-8+6)$$

$$= -40 + 108 - 20 + 15 + 12$$

$$c. \begin{vmatrix} 4 & 2 & 4 & -1 & -2 & 0 \\ & -1 & -3 & & 2 & 4 \end{vmatrix}$$

$$= 4(-6+4) - (-8-0)$$

$$= -8+8$$

$$= 0$$

$$7. a. \begin{bmatrix} \begin{vmatrix} 4 & 1 \\ -4 & 5 \end{vmatrix} & -\begin{vmatrix} -3 & 1 \\ 4 & 5 \end{vmatrix} & \begin{vmatrix} -3 & 4 \\ 4 & -4 \end{vmatrix} \\ -\begin{vmatrix} 2 & 8 \\ -4 & 5 \end{vmatrix} & \begin{vmatrix} 6 & 8 \\ 4 & 5 \end{vmatrix} & -\begin{vmatrix} 6 & 2 \\ 4 & -4 \end{vmatrix} \\ \begin{vmatrix} 2 & 8 \\ 4 & 1 \end{vmatrix} & \begin{vmatrix} 6 & 8 \\ -3 & 1 \end{vmatrix} & \begin{vmatrix} 6 & 2 \\ -3 & 4 \end{vmatrix} \end{bmatrix}^T = \begin{bmatrix} 24 & 19 & -4 \\ -42 & -2 & 32 \\ -30 & -30 & 30 \end{bmatrix}^T = \begin{bmatrix} 24 & -42 & -30 \\ 19 & -2 & -30 \\ -4 & 32 & 30 \end{bmatrix}$$

$$b. 6 \begin{vmatrix} 4 & 1 \\ -4 & 5 \end{vmatrix} - 2 \begin{vmatrix} -3 & 1 \\ 4 & 5 \end{vmatrix} + 8 \begin{vmatrix} -3 & 4 \\ 4 & -4 \end{vmatrix} = 6(24) - 2(-42) + 8(-30) = 150$$

$$c. \begin{bmatrix} 6 & 2 & 8 \\ -3 & 4 & 1 \\ 4 & -4 & 5 \end{bmatrix} \begin{bmatrix} 24 & -42 & -30 \\ 19 & -2 & -30 \\ -4 & 32 & 30 \end{bmatrix} = \begin{bmatrix} 150 & 0 & 0 \\ 0 & 150 & 0 \\ 0 & 0 & 150 \end{bmatrix} = 150 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

8. Since $a_{ij} \in \mathbb{Z}$, then the minor of a_{ij} which is $\det(M_{ij})$ must also be an integer. Thus, the cofactor $(a_{ij}) = (-1)^{i+j} \det(M_{ij})$ is also an integer, and the elements of $\text{adj } A = (\text{cofactor}(a_{ij}))^T$ must also be integers. Then $A^{-1} = (\text{adj } A) / (\det A) = \pm(\text{adj } A)$ must also be integers.

a. Only if square.

$$\det(AB) = \det(A)\det(B)$$

$$= \det(B)\det(A) = \det(BA)$$

$$A = \begin{bmatrix} 1 & 2 \end{bmatrix} \quad BA = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 4 \end{bmatrix} \quad \det(BA) = 0 \quad \left. \begin{array}{l} \det(AB) = 4 \\ \det(BA) = 0 \end{array} \right\} \begin{array}{l} \text{not} \\ \text{equal} \end{array}$$

$$10. \det(2A) = 2^4 \det(A) = 16 \cdot \frac{1}{2} = 8$$

$$\det(-A) = (-1)^4 \det(A) = 1 \cdot \frac{1}{2} = \frac{1}{2}$$

$$\det(A^2) = \det(A)\det(A) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$\det(A^{-1}) = 1/\det(A) = 1/(1/2) = 2$$